10.6. What is the *Q*-value for the reactions: (a) $^{11}B(d,\alpha)^{9}Be$; (b) $^{7}Li(p,n)^{7}Be$?

10.7. What is the maximum velocity that a deuteron of 2 MeV can impart to a ^{16}O atom?

10.8. Calculate the mass of an electron accelerated through a potential of 2×10^{8} V.

10.9. ^{12}C atoms are used to irradiate ^{239}Pu to produce an isotope of berkelium. What is the Coulomb barrier height?

10.10. Measurements made on the products of the reaction $^{7}Li(d,\alpha)^{5}He$ have led to an isotopic mass of 5.0122 for the hypothetical nuclide ^{5}He. Show that this nuclear configuration cannot be stable by considering the reaction $^{5}He \rightarrow {}^{4}He + n$.

10.11. In an experiment one hopes to produce the long-lived (2.6 y) ^{22}Na through a d,2n-reaction on neon. What is (a) the *Q*-value, (b) the threshold energy, (c) the Coulomb barrier height, and (d) the minimum deuteron energy for the reaction? The mass excesses (in keV) are −5185 for ^{22}Na and −8027 for ^{22}Ne.

10.11. LITERATURE

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CHAPTER 11

Mechanisms and Models of Nuclear Reactions

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The variety and complexity of nuclear reactions make this a fascinating area of research quite apart from the practical value of understanding fusion and fission. From studies of such properties as the relative amounts of formation of various competing products, the variation of the yields of these with bombarding energy, the directional characteristics and kinetic energies of the products, etc., we may formulate models of nuclear reaction mechanisms. Such models lead to systematics for nuclear reactions and make possible predictions of reactions not yet investigated. As such, the nuclear reactions can be simulated in fashons very similar to those used for thermodynamic modelling of chemical reactions where the yield of different species are governed by stability constants which in turn may be differently affected by, e.g. changes in temperature etc.

11.1. THE REACTION CROSS-SECTION

The probability for a nuclear reaction to occur is expressed in terms of the *reaction cross-section*. The geometric cross-section that a nucleus presents to a beam of particles is πr^2. If

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